A MAP FROM A FINITE FREE MODULE LANDING IN THE IMAGE OF ANOTHER LIFTS ALONG IT

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Allowed standard prerequisites (editorial).

- A basis of the finite free module. Used at the first line of the counted proof (line 600). Reduces the construction of the lifting map to a choice on the finitely many basis vectors.
- Preimages under a surjection onto the image. The hypothesis that the image of alpha lies in the image of beta, used at proof lines 601-602. Supplies, for each basis vector, an element of N mapping to its image under alpha.

Lemma 0.1. *Let R be a ring. Let $\alpha : R^{\oplus n} \to M$ and $\beta : N \to M$ be module maps. If $\mathrm{Im}(\alpha) \subset \mathrm{Im}(\beta)$, then there exists an R-module map $\gamma : R^{\oplus n} \to N$ such that $\alpha = \beta \circ \gamma$.*

Proof. Let $e_i = (0, \ldots, 0, 1, 0, \ldots, 0)$ be the ith basis vector of $R^{\oplus n}$. Let $x_i \in N$ be an element with $\alpha(e_i) = \beta(x_i)$ which exists by assumption. Set $\gamma(a_1, \ldots, a_n) = \sum a_i x_i$. By construction $\alpha = \beta \circ \gamma$. □

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